

# Field Manual & Operations Guide

Complete reference for setup, administration and day-to-day business operations.

---

## CONTENTS

1. System Overview
2. Getting Started
3. Admin Panel
4. Storefront Guide
5. Business Operations
6. Technical Reference

SECTION 1

# System Overview

What BIZShop is and how it works

## About This System

BIZShop is a self-hosted ecommerce platform built for Australian small businesses selling tools, 3D printed parts, electronics components, and hardware. It runs entirely on your own server — no monthly subscription or third-party ecommerce platform required.

## Key Features

- Full product catalogue — 10 categories, 74+ products out of the box
- Shopping cart and multi-step checkout with Australian GST (10%) calculated automatically
- Flexible shipping: always free, flat rate, or individual rates per state/territory
- Customer order self-tracking by order number and email — no account required
- Admin panel: products, orders, inventory, customers, activity log, and settings
- Low stock badges on the storefront and email alerts when stock hits your threshold
- Packing slips and ATO-compliant tax invoices with ABN printed automatically
- Shop filtering by price range, in-stock status, category, and sort order

## Technology Stack

| Component       | Technology                       | Notes                     |
|-----------------|----------------------------------|---------------------------|
| Backend runtime | Node.js                          | v18+ (v24 recommended)    |
| Web framework   | Express.js                       | v4.18                     |
| Database        | JSON flat-file (store.json)      | No external DB required   |
| File uploads    | Multer                           | v2 — 5 MB image limit     |
| Email           | Nodemailer                       | v8 — requires SMTP config |
| Frontend        | Vanilla JS SPA                   | History API routing       |
| Charts          | Chart.js                         | v4.4 via CDN              |
| Authentication  | Session tokens + HttpOnly cookie | 8-hour TTL                |

**NOTE**

All business data — products, orders, customers, settings, and activity log — is stored in a single file: store.json. Back this file up regularly. Uploaded product images are stored separately in public/uploads/.

## SECTION 2

# Getting Started

Installation, first run, and initial configuration

## System Requirements

- Node.js v18 or later — download from [nodejs.org](https://nodejs.org) (v24 LTS recommended)
- Windows 10/11, macOS, or Linux
- At least 512 MB RAM, 1 GB recommended
- Port 3000 free, or set a custom PORT environment variable

## Installation

### Step 1 — Install dependencies

Open a terminal in the project folder (C:\Users\Administrator\Desktop\BIZ) and run:

```
npm install
```

### Step 2 — Start the server

```
node server.js

# Development mode (auto-restarts on file changes):
npx nodemon server.js
```

### Step 3 — Open the store

Once started, your terminal will show:

```
Store running at http://localhost:3000
Admin panel at http://localhost:3000/admin
```

Other devices on the same local network can reach the store using your PC's IP address instead of localhost — e.g. <http://192.168.1.50:3000>.

## Admin Login

Navigate to <http://localhost:3000/admin>. You will be redirected to the login page. Enter the admin password stored in `auth.js` (line 3).

**WARNING**

The admin password is stored as plain text in `auth.js`. Change it before exposing the server to the public internet. Restart the server after changing it.

## First-Time Setup Checklist

Complete these steps in the Admin Panel !' Settings before going live:

- Store Name, Tagline, and ABN — the ABN is legally required on Tax Invoices
- Contact Email and Phone — displayed on packing slips and invoices
- Street Address — your business address printed on documents
- Shipping mode — choose Free, Flat Rate, or By State/Territory
- Low Stock Threshold — alerts fire when any product hits this level (default: 5)
- SMTP credentials — required only if you want stock alert emails
- Review product prices, stock counts, and SKUs for accuracy
- Place a full test order end-to-end to confirm checkout and fulfilment work correctly

**TIP**

Use a real email address you control for the test order. Walk the order through every status in the admin panel — from Pending through to Delivered — to confirm the entire workflow.

## SECTION 3

# Admin Panel

Complete guide to every section of the administration interface

## Accessing the Admin Panel

URL: <http://localhost:3000/admin> (replace localhost with your server IP or domain when deployed). Admin sessions last 8 hours — you are automatically returned to the login page after the session expires.

### 3.1 Dashboard

The first screen after login. A real-time summary of the business at a glance.

#### Stat Cards

- Active Items — products currently visible in the storefront
- Categories — total number of product categories
- Total Orders — all orders ever placed
- Pending Orders — orders awaiting action, shown in amber as a priority flag
- Revenue — sum of all non-cancelled orders including GST and shipping

#### Charts

- Revenue — Last 30 Days: a daily bar chart. Hover over any bar to see the exact figure.
- Orders by Status: a doughnut showing the proportion of orders in each status.
- Top Products by Revenue: a horizontal bar ranking your highest-earning products.

#### Low Stock Alert Table

Lists every product at or below the configured Low Stock Threshold. Use this to initiate reorders before stock runs out. Click any row to jump to the Inventory section for that product.

### 3.2 Categories

Manages the top-level groupings shown in the shop sidebar, homepage grid, and product breadcrumbs.

#### Creating a Category

- Click "+ New Category"
- Enter a Name (required) and an optional Description
- Save — the URL slug is generated automatically from the name (e.g. "Hand Tools" !' hand-tools)

#### Editing

- Click "Edit" on any row, update the fields, and save.

#### Deleting

- Only permitted when the category contains zero products. Reassign or delete all products first.

#### NOTE

Changing a category name also changes its URL slug. Any shared links or saved bookmarks that used the old slug URL will stop working.

### 3.3 Items (Products)

The full product catalogue. Use the dropdowns at the top to filter by category, or show hidden (inactive) products.

#### Adding a New Product

- Click "+ New Item" to open the product modal
- Select the Category
- Enter Name, Description, Price (ex-GST — tax is added at checkout), Initial Stock, and SKU
- Upload an image by dragging onto the image zone or clicking Browse (JPEG/PNG/WEBP, max 5 MB)
- Tick "Active" to make the product visible immediately
- Tick "Featured" to highlight it on the homepage
- Click "Save Item"

#### Editing

- Click "Edit" on any row. All fields including the image can be changed. To remove the current image, click the ' on the preview before saving.

#### Hiding vs. Deleting

- Deactivate — untick Active in the edit modal. Product disappears from the store but all data is preserved. Can be reactivated at any time.
- Delete — permanently removes the product and all its inventory history. This cannot be undone.

##### WARNING

Deleting a product that exists in historical orders does not corrupt those orders — item names and prices are snapshotted at purchase time. But the product can no longer be managed or reactivated.

### 3.4 Inventory

Manually adjust stock levels and view the permanent adjustment history for any product.

#### Making a Stock Adjustment

- Select a product from the dropdown at the top
- Enter the change amount — positive to add stock (e.g. +50 for a new delivery), negative to reduce it (e.g. -2 for damaged goods)
- Enter a meaningful Reason — this is recorded permanently in the audit log
- Click Apply

#### Adjustment Log

Every change is permanently recorded with date/time, the change amount, resulting stock level, and reason. This log cannot be edited — it is your inventory audit trail.

##### TIP

Always write a detailed reason. "Received from Jaycar — Invoice #4821 — 2026-05-19" will be far more useful than "new stock" when reviewing records months later.

## 3.5 Orders

All customer orders, filterable by status via the tab row. Click any order row to open its full detail.

### Order Status Workflow

| Status           | Meaning                               | Suggested next action               |
|------------------|---------------------------------------|-------------------------------------|
| Pending          | Placed — payment not yet confirmed    | Confirm payment, then !' Processing |
| Processing       | Payment confirmed — picking & packing | Pack and dispatch !' Shipped        |
| Shipped          | Dispatched — in transit               | Await delivery confirmation         |
| Delivered        | Customer has received the order       | No further action required          |
| Cancelled        | Order voided before fulfilment        | Issue refund if payment was taken   |
| Return Requested | Customer has requested a return       | Assess claim, then approve/reject   |
| Refunded         | Return processed, money returned      | No further action required          |

### Processing an Order — Step by Step

- Click the order row to open the detail modal
- Read customer details, shipping address, and any order notes
- Add internal notes in the "Internal Notes" field (not visible to the customer)
- Change status to "Processing" once payment is confirmed
- Print the Packing Slip to use as a pick-and-pack reference in the warehouse
- After dispatching, change status to "Shipped" and note the tracking number in Internal Notes
- Change to "Delivered" once confirmed, or after the expected delivery window has passed

### Packing Slips & Tax Invoices

- Packing Slip — items, quantities, and delivery address. For warehouse use.
- Tax Invoice — full GST-compliant invoice with ABN, itemised lines, GST total, and grand total. For the customer's records.

#### NOTE

Your ABN must be entered in Settings for it to appear on Tax Invoices. Invoices missing an ABN may not meet ATO requirements for GST credit claims.

### Exporting Orders to CSV

Click "+ Export CSV" in the Orders section header. The export respects the active status filter tab, so you can export only pending orders, only shipped orders, or all at once. The CSV is compatible with Excel, Xero, and MYOB.

## 3.6 Customers

A read-only customer list built automatically from order history. No customer registration is required.

- Search by name or email using the search bar
- Table shows: name, email, phone, state, order count, total spent, last order date

- Click any row to see a full customer profile with order history
- Click any order in the customer profile to open that order's detail modal

### 3.7 Activity Log

A chronological audit trail of every significant action in the system. Populated automatically — nothing to configure.

| Event type | Icon | What triggers it                                  |
|------------|------|---------------------------------------------------|
| Order      | 📄    | New order placed; any order status change         |
| Item       | 📦    | Product created, updated, deactivated, or deleted |
| Inventory  | 📊    | Stock level manually adjusted                     |
| Category   | 📁    | Category created, updated, or deleted             |
| Settings   | ⚙️   | Any settings field saved                          |
| Auth       | 👤    | Admin signed in                                   |

Filter by event type using the tabs. Timestamps display as "just now", "3m ago", "2h ago", then a full date for older entries. The log retains the 500 most recent entries.

### 3.8 Settings

Store-wide configuration. Click "Save Settings" after making any changes.

#### Store Identity

- Store Name — appears in browser tabs, emails, and invoices
- Tagline — short descriptor shown under the logo
- ABN — Australian Business Number, printed on all Tax Invoices

#### Shipping Options

- Always Free — no charge on any order
- Flat Rate — a single dollar amount for all orders. Optionally set a Free Shipping Threshold (e.g. A\$100) above which shipping is waived.
- By State/Territory — set a different rate for each of the 8 Australian states/territories. The flat rate acts as a fallback for any state not individually configured.

#### Stock Alerts

- Low Stock Threshold — products at or below this level show a badge in the store and trigger an email alert (if SMTP is configured)
- Alert Email — address for alert notifications; defaults to Contact Email if left blank

#### Email / SMTP Configuration

| Field     | Gmail example  | Notes                             |
|-----------|----------------|-----------------------------------|
| SMTP Host | smtp.gmail.com |                                   |
| Port      | 587            | Use 465 for SSL, 587 for STARTTLS |



|          |                         |                                  |
|----------|-------------------------|----------------------------------|
| Username | you@gmail.com           | Your full email address          |
| Password | App password (16 chars) | NOT your regular Google password |

**NOTE**

Gmail requires an App Password. Enable 2-Step Verification in your Google Account, then go to Security !' App passwords !' generate one for "Mail". Paste the 16-character code into the Password field — not your regular Gmail password.

## SECTION 4

# Storefront Guide

The customer-facing website — every page explained

## 4.1 Homepage ( / )

- Hero — store name, tagline, and "Shop Now" button
- Category cards — one per category showing product count; clicking navigates to that category in the shop
- Popular Products — the first 8 active products shown as cards with price, stock badge, and Add to Cart

## 4.2 Shop ( /shop )

### Sidebar — desktop only

- Categories list — "All Products" or any individual category
- In stock only checkbox — hides out-of-stock products instantly
- Price Range — Min and Max inputs; filter applies 300ms after you stop typing

### Top bar

- Heading — current category name, search query, or "All Products"
- Item count — updates in real-time as filters are applied
- Sort dropdown — Default / Price: Low!'High / Price: High!'Low / Name: A!'Z / Name: Z!'A

#### NOTE

On mobile, the sidebar is hidden. The sort dropdown remains visible. Customers can browse by category from the navigation menu or footer links.

## 4.3 Product Detail ( /product/:id )

- Breadcrumb — Home / Shop / Category / Product Name
- Product image — click to open a fullscreen lightbox
- Stock badge — "In Stock", "Only X left" (low stock in amber), or "Out of Stock"
- Quantity picker — capped at current available stock
- Add to Cart — disabled and labelled "Sold Out" when stock is zero
- Related products — up to 4 products from the same category

## 4.4 Cart ( /cart )

Shows current cart items stored in browser localStorage — persists across browser sessions on the same device.

- Adjust quantity or remove individual items
- Live summary — subtotal, GST (10%), and estimated shipping
- "Proceed to Checkout" button

**NOTE**

Cart data lives in the customer's browser only. It is not visible in the admin panel. Clearing browser data or switching devices starts with an empty cart.

## 4.5 Checkout ( /checkout )

- Required: Full Name, Email, Street Address, City, State/Territory, Postcode
- Optional: Phone Number, Order Notes
- Live order summary: shipping cost updates automatically when State is selected
- On completion: stock deducted, customer redirected to Order Confirmation with their order number

## 4.6 Order Tracking ( /track )

Customers enter the email address used at checkout and their Order Number (format: BIZ-XXXXXX). The result shows status, items ordered, totals, and shipping address — no account required.

## 4.7 Search

The header search bar is active on every page. Typing shows a live dropdown of up to 8 matching products by name or description. Pressing Enter navigates to the shop page filtered by the search query.

## SECTION 5

# Business Operations

Day-to-day workflows and Australian compliance

## 5.1 Daily Opening Routine

- Open the Admin Dashboard — check for overnight or weekend orders
- Review the Pending tab in Orders — process any new confirmed payments
- Check the Low Stock Alert table — initiate reorders for items running low
- Scan the Activity Log for any unexpected changes

## 5.2 Full Order Processing Workflow

1. Open the order in the Orders section and read the customer's details and notes
2. Confirm payment via your payment method (bank transfer, Square, Stripe, etc.)
3. Change status to 'Processing'
4. Print the Packing Slip — use it to pick and pack the items
5. Dispatch via Australia Post or your preferred carrier
6. Change status to 'Shipped'. Record the tracking number in Internal Notes.
7. Customer can now see "Shipped" on the order tracking page
8. Change status to 'Delivered' after confirmation or after the expected delivery window

**WARNING**

BIZShop does not process payments — it is a catalogue and order management system only. All payment collection and refund processing must be handled through your bank, Square, Stripe, or another payment method that you advertise to customers.

## 5.3 Handling Returns & Refunds

- Customer contacts you requesting a return — update order status to "Return Requested"
- Assess the claim against your returns policy
- If approved, arrange return postage with the customer
- Once goods are received and inspected, process the refund through your payment method
- Update order status to "Refunded" and document the resolution in Internal Notes

## 5.4 Managing Stock

### Receiving a new delivery

- Go to Inventory — select the product
- Enter a positive adjustment equal to the quantity received (e.g. +100)
- Reason: "Received from [Supplier] — Inv #XXXX — [Date]"
- Click Apply

### Writing off damaged or lost stock

- Select the product in Inventory

- Enter a negative adjustment (e.g. -3)
- Reason: "Damaged in transit" or "Missing at stock count — [Date]"
- Click Apply

### Periodic stock takes

Compare physical counts against the Current Stock figure shown in the Inventory section. For each discrepancy, apply an adjustment with reason "Stock take correction — [Date]".

## 5.5 Australian GST & Tax Invoices

GST is automatically calculated at 10% on the product subtotal. Shipping is shown separately and does not attract additional GST on top.

### ATO Tax Invoice requirements — all included automatically

- The words "Tax Invoice" '
- Your business name (from Settings !' Store Name) '
- Your ABN (from Settings !' ABN — must be filled in) '
- Date of issue (the order date) '
- Description of goods supplied (itemised product list) '
- The GST amount payable '

#### NOTE

For supplies over A\$1,000, the invoice must also show the customer's name and address. The BIZShop Tax Invoice includes both fields automatically from the order record.

## 5.6 Pricing Notes

- All prices are entered ex-GST. GST (10%) is added at checkout — not stored in the product price.
- The Revenue stat on the Dashboard includes GST and shipping in the total.
- Review "Top Products by Revenue" monthly to prioritise restocking of best-sellers.
- For component kits, price them to reflect a genuine saving over buying items individually.

## SECTION 6

# Technical Reference

File structure, API reference, and maintenance procedures

## 6.1 File & Folder Structure

```

BIZ/
  server.js           Main application entry point
  auth.js            Session management & admin password
  database.js        All data read/write logic
  mailer.js          Email via Nodemailer (SMTP)
  store.json          YOUR DATABASE – back this up!
  package.json        Node.js dependencies
  routes/
    orders.js         Order API
    items.js          Product API (incl. image upload)
    categories.js     Category API
    settings.js       Settings API
    activity.js        Activity log API
  public/
    index.html        Storefront SPA shell
    admin.html        Admin panel shell
    login.html        Admin login page
    css/style.css     All styles – storefront + admin
    js/app.js         Storefront SPA logic
    js/admin.js       Admin panel logic
    uploads/          Uploaded product images
  
```

## 6.2 API Endpoint Reference

| Method | Endpoint                 | Auth | Purpose                                |
|--------|--------------------------|------|----------------------------------------|
| GET    | /api/categories          | No   | List all categories with item counts   |
| POST   | /api/categories          | Yes  | Create a new category                  |
| PUT    | /api/categories/:id      | Yes  | Update a category                      |
| DELETE | /api/categories/:id      | Yes  | Delete a category (only if empty)      |
| GET    | /api/items               | No   | List products (filterable)             |
| POST   | /api/items               | Yes  | Create a product (multipart/form-data) |
| PUT    | /api/items/:id           | Yes  | Update a product                       |
| DELETE | /api/items/:id           | Yes  | Deactivate or permanently delete       |
| PATCH  | /api/items/:id/inventory | Yes  | Adjust stock level                     |

|       |                             |     |                                    |
|-------|-----------------------------|-----|------------------------------------|
| POST  | /api/orders                 | No  | Place a new order (checkout)       |
| GET   | /api/orders                 | Yes | List all orders                    |
| GET   | /api/orders/lookup          | No  | Customer lookup by email + order # |
| PATCH | /api/orders/:id/status      | Yes | Update order status                |
| PATCH | /api/orders/:id/admin-notes | Yes | Save internal admin notes          |
| GET   | /api/settings               | No  | Get store settings                 |
| PUT   | /api/settings               | Yes | Save settings                      |
| GET   | /api/activity               | Yes | Activity log (up to 500 entries)   |

## 6.3 Changing the Admin Password

Open `auth.js` and update line 3, then restart the server:

```
// auth.js line 3
const ADMIN_PASSWORD = 'YourNewSecurePassword!';

// Restart the server:
node server.js
```

### WARNING

Changing the password does not immediately invalidate active admin sessions — they remain valid until their 8-hour TTL expires. To instantly log out all sessions, restart the server.

## 6.4 Backup Procedure

### What to back up

- `store.json` — your entire database (orders, products, settings, activity log)
- `public/uploads/` — all uploaded product images
- `auth.js` — contains your admin password

### Manual backup — Windows PowerShell

```
$date = Get-Date -Format "yyyyMMdd"
Copy-Item store.json "store-backup-$date.json"
```

### Manual backup — Mac / Linux

```
cp store.json store-$(date +%Y%m%d).json
```

**TIP**

Keep at least 7 days of rolling backups stored on a separate drive or cloud service (OneDrive, Google Drive) so that a single disk failure cannot wipe both your live data and the backup.

## 6.5 Keeping the Server Running

The server stops if the terminal is closed or the PC reboots. Use PM2 to keep it running as a background service that survives reboots:

```
npm install -g pm2
pm2 start server.js --name bizshop
pm2 save
pm2 startup    # follow the printed instruction to enable boot auto-start

# Useful PM2 commands:
pm2 status
pm2 restart bizshop
pm2 logs bizshop
```

## 6.6 Troubleshooting

| Problem                     | Likely cause                 | Fix                                                     |
|-----------------------------|------------------------------|---------------------------------------------------------|
| Port 3000 already in use    | Another process running      | Kill the other process or set PORT=3001                 |
| Admin password not accepted | Wrong value in auth.js       | Check ADMIN_PASSWORD in auth.js; restart server         |
| Product images not showing  | Upload folder missing        | Check public/uploads/ exists; ensure server runs        |
| Alert emails not sending    | SMTP config incorrect        | Verify host, port, username, app password in Settings   |
| All store data missing      | store.json deleted/corrupted | Restore from backup — the file is the entire database   |
| Blank white screen          | JS error in browser          | F12 !' Console tab — read the error message             |
| Order total incorrect       | Shipping/GST misconfigured   | Review shipping settings; GST is always 10% of subtotal |



# BIZShop

## Field Manual & Operations Guide

---

Tools & 3D Printed Parts · Electronics · Hardware

For internal use only. Keep this document accessible to all staff who manage the store.